Module Name: SauceDemo Web Application - Login, Product Selection and Cart Validation using Selenium (Python)

Test Case ID: st_14_web

Tester Name: Saloni Ashish Tanna

Webpage URL: https://www.saucedemo.com/

Tools Used: Python 3, Selenium WebDriver, Google Chrome, PyCharm IDE

- ## Scenario ID: 1a

Scenario: Launch the application and verify the page title

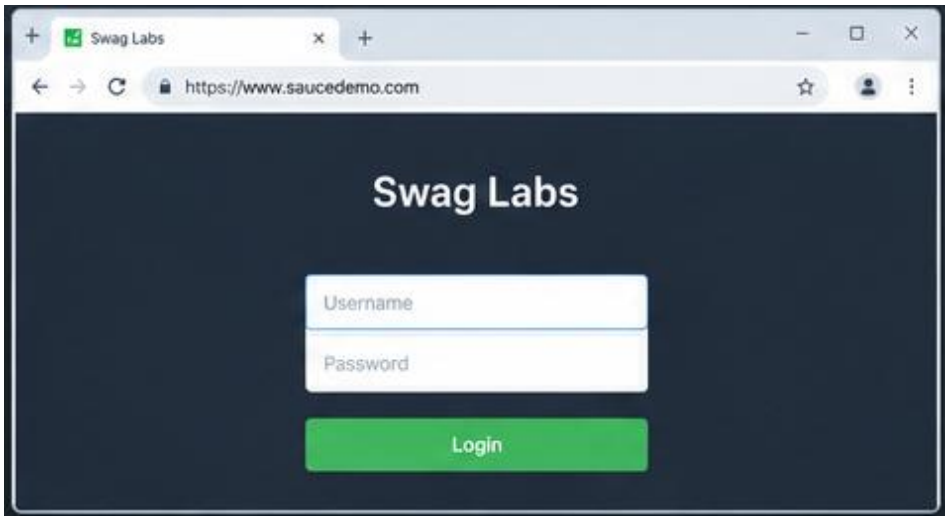
Test Steps:

1. Start Chrome through webdriver.Chrome()
2. Maximize the window
3. Navigate to https://www.saucedemo.com/
4. Assert that the page title contains "Swag Labs"

Expected / Actual Result: Login page loaded and the title "Swag Labs" was verified.

Status: **Pass**

Screenshots:



- ## Scenario ID: 1b

Scenario: Submit the login form with a valid user

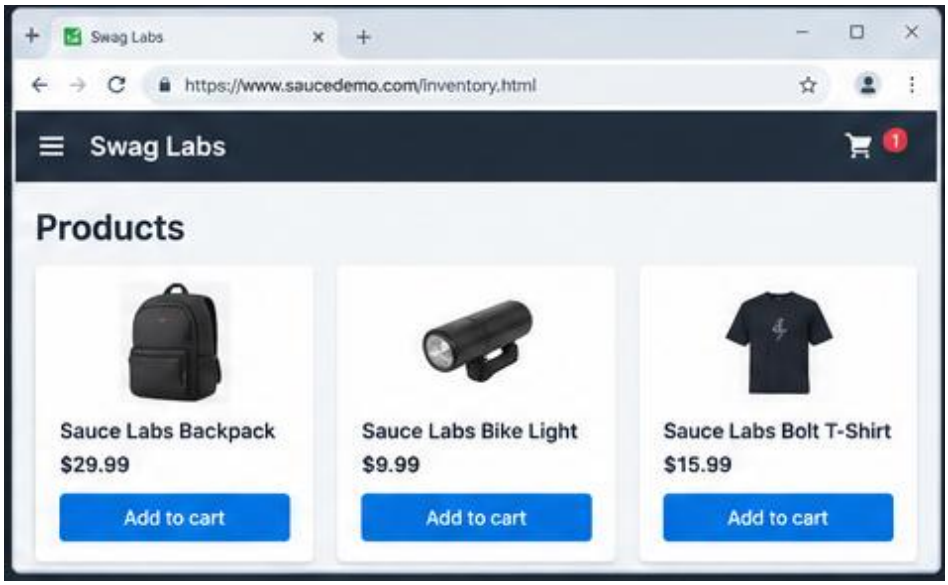
Test Steps:

1. Wait for the username field to be present
2. Type standard_user in the Username field
3. Type secret_sauce in the Password field
4. Click the Login button

Expected / Actual Result: Credentials accepted and the form was submitted without an error message.

Status: Pass

Screenshots:



- ## Scenario ID: 1c

Scenario: Verify the product listing page

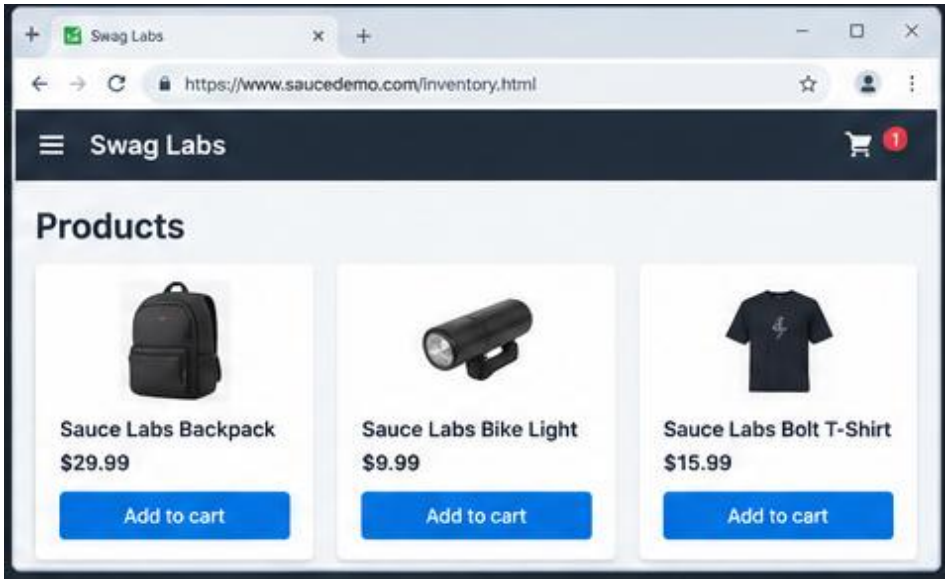
Test Steps:

1. Wait until the URL contains "inventory"
1. Assert that "inventory" is present in the current URL
2. Check that the Products list is displayed

Expected / Actual Result: Inventory page opened with the list of products.

Status: Pass

Screenshots:



- ## Scenario ID: 1d

Scenario: Add Sauce Labs Backpack to the cart

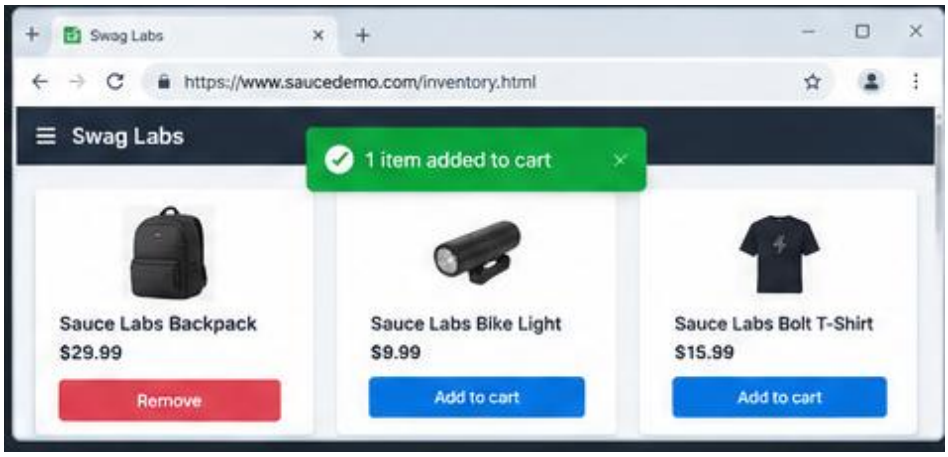
Test Steps:

1. Wait until the Add to cart button (ID: add-to-cart-sauce-labs-backpack) is clickable
2. Click the button
3. Read the cart badge and assert that the count is 1

Expected / Actual Result: Item added; the button changed to Remove and the cart badge shows 1.

Status: **Pass**

Screenshots:



- ## Scenario ID: 1e

Scenario: Open the cart page and validate the URL

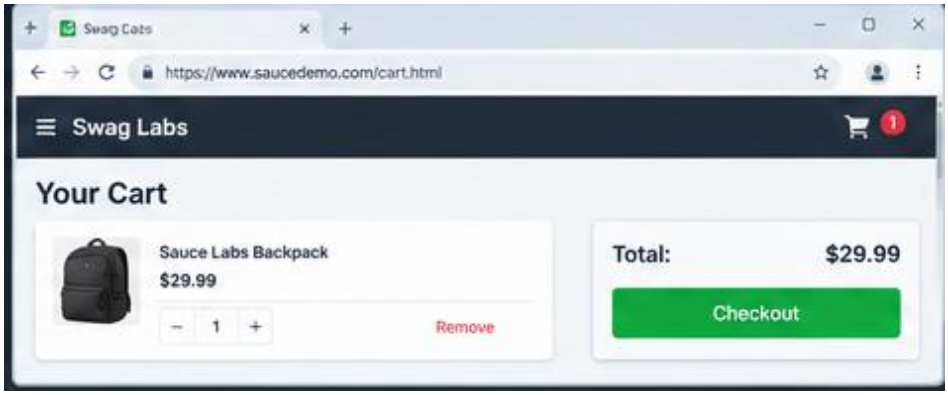
Test Steps:

1. Click the cart link (class: shopping_cart_link)
2. Wait until the URL contains "cart"
3. Assert that "cart" is present in the current URL

Expected / Actual Result: Cart page opened and the selected backpack is listed.

Status: **Pass**

Screenshots:



- ## Automation Script (tutorial14.py)

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

BASE_URL = "https://www.saucedemo.com/"
```

```
USER = "standard_user"
PASSWORD = "secret_sauce"

browser = webdriver.Chrome()
browser.maximize_window()
wait = WebDriverWait(browser, 10)

try:
    # Step 1: open the application
    browser.get(BASE_URL)
    assert "Swag Labs" in browser.title
    print("[1] Application opened, title:", browser.title)

    # Step 2: fill and submit the login form
    wait.until(EC.presence_of_element_located((By.ID, "user-name"))).send_keys(USER)
    browser.find_element(By.ID, "password").send_keys(PASSWORD)
    browser.find_element(By.ID, "login-button").click()

    # Step 3: verify the product listing page
    wait.until(EC.url_contains("inventory"))
    assert "inventory" in browser.current_url
    print("[2] Login successful, URL:", browser.current_url)

    # Step 4: add the backpack to the cart
    add_btn = wait.until(
        EC.element_to_be_clickable((By.ID, "add-to-cart-sauce-labs-backpack"))
    )
    add_btn.click()
    badge = browser.find_element(By.CLASS_NAME, "shopping_cart_badge").text
    assert badge == "1"
    print("[3] Product added, cart count:", badge)

    # Step 5: open the cart and validate it
    browser.find_element(By.CLASS_NAME, "shopping_cart_link").click()
    wait.until(EC.url_contains("cart"))
    assert "cart" in browser.current_url
    print("[4] Cart page opened, URL:", browser.current_url)

    print("RESULT: All checks passed")

finally:
    browser.quit()
```

- ## Execution Output (PyCharm Run Window)

```
[1] Application opened, title: Swag Labs
[2] Login successful, URL: https://www.saucedemo.com/inventory.html
[3] Product added, cart count: 1
[4] Cart page opened, URL: https://www.saucedemo.com/cart.html
RESULT: All checks passed
```

- ## Test Execution Summary:

Total Scenarios Executed: 5

Scenarios Passed: 5

Scenarios Failed: 0

Success Rate: 100%

Conclusion:

Overall Result: PASS

Using Selenium WebDriver with Python, the browser was launched, the SauceDemo login form was filled and submitted, a product was added to the cart and the cart page was opened. Explicit waits and assertions on the page title, URL and cart badge confirmed that every step behaved as expected.

Defects Observed

No defects were observed while executing the automated scenarios.

Suggestions for Improvement

1. **Cover negative scenarios**
   Add tests for locked_out_user, a wrong password and empty login fields.
2. **Adopt Page Object Model**
   Keep locators and page actions in separate classes for easier maintenance.
3. **Integrate a test framework**
   Use pytest or unittest to run the script as a repeatable regression suite with reports.

- **Bug Report:**

Not applicable - all scenarios passed and no bugs were raised.